ELEVANCESKILLS

Embedded Systems Internship

PROJECT REPORT

Battery Intelligence System using ESP32 and Blynk IoT

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Abstract

The Battery Intelligence System is an IoT-based Smart Battery Management System (BMS) developed using the ESP32 microcontroller, Blynk Cloud platform, and Wokwi simulator. The project continuously monitors a simulated four-cell lithium battery pack by measuring individual cell voltages, calculating pack voltage, battery health, imbalance percentage, and identifying the weakest and strongest cells.

The system also implements an intelligent safety protection mechanism capable of detecting weak cells, overvoltage conditions, sensor failures, frozen ADC values, relay mismatches, and rapid voltage fluctuations. During abnormal operating conditions, the controller automatically activates the relay protection circuit and buzzer while displaying warning messages on the LCD.

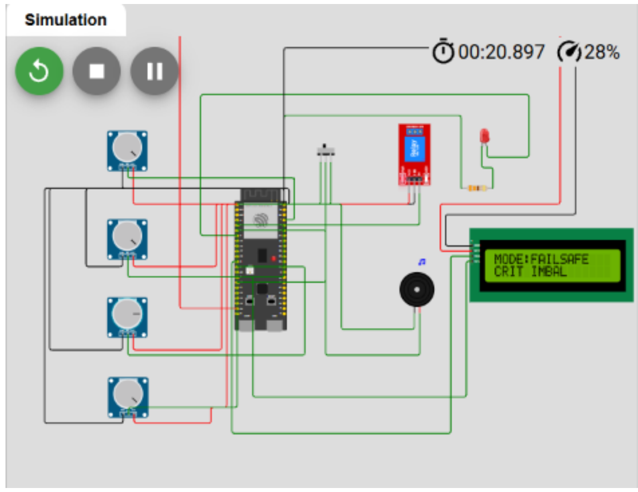
Cloud connectivity is achieved using the Blynk IoT platform, allowing real-time monitoring of battery parameters, fault history, operator recommendations, battery state of charge (SOC), risk level, and voltage trends. The project demonstrates the practical implementation of embedded systems, IoT communication, real-time monitoring, and intelligent fault management for battery-powered applications.

Introduction

Battery Management Systems (BMS) play an essential role in electric vehicles, renewable energy storage, robotics, and portable electronic devices. They ensure safe battery operation by continuously monitoring voltage levels, detecting abnormal conditions, and protecting the battery from potential damage.

In this project, a smart Battery Intelligence System was designed using the ESP32 microcontroller. The system monitors four battery cells in real time and performs battery analytics, fault detection, runtime management, and cloud telemetry. The integration of the Blynk IoT platform enables remote monitoring through an interactive dashboard, providing users with live battery information and intelligent recommendations.

The project demonstrates the importance of embedded systems in modern battery monitoring applications and showcases practical implementation using simulation before deployment on real hardware.



Objectives

• Real-time battery monitoring
• Battery health classification
• Fault detection and protection
• LCD HMI
• Cloud telemetry using Blynk
• Runtime modes and recovery logic

Hardware & Software

Hardware: ESP32-S3, LCD 16x2 I2C, Relay, Buzzer, Potentiometers.
Software: Arduino IDE, Wokwi Simulator, Blynk IoT, GitHub.

System Features

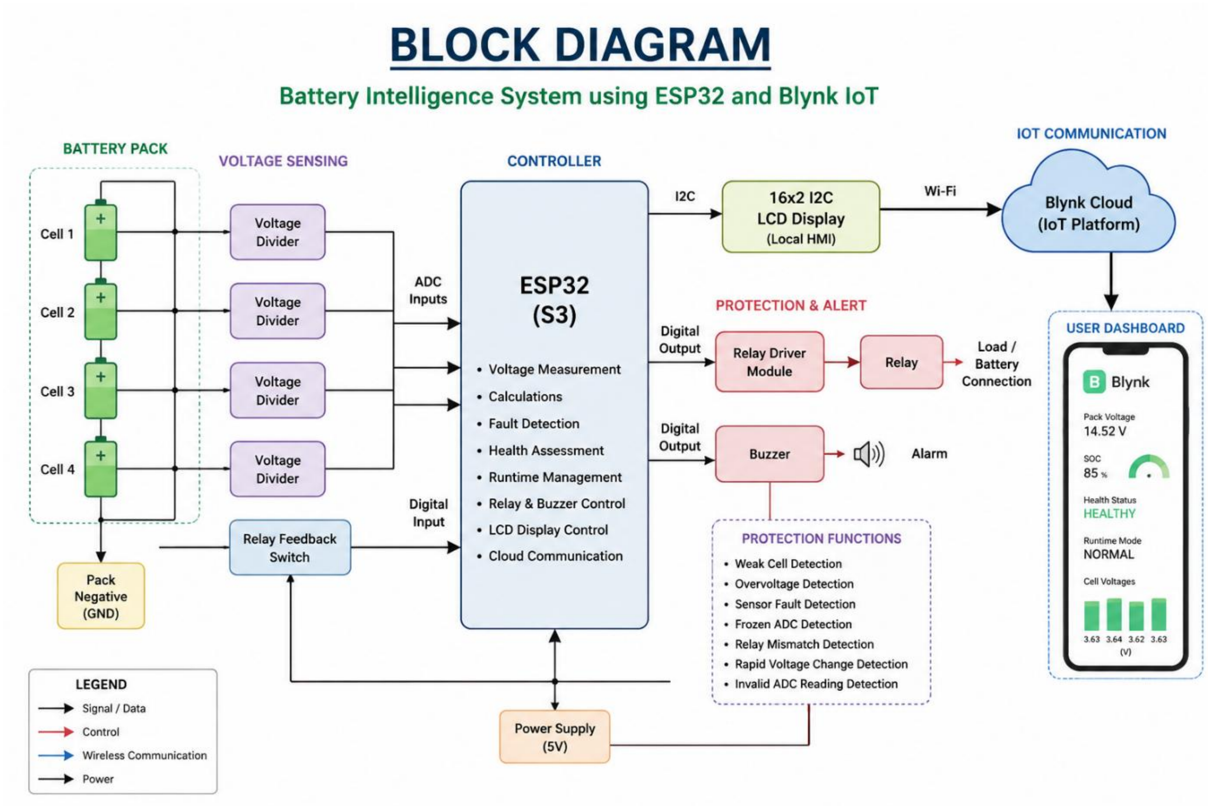
The developed Battery Intelligence System includes the following features:

- Real-time monitoring of four battery cells.
- Automatic calculation of pack voltage.
- Average voltage computation.
- Battery imbalance percentage calculation.
- Weakest and strongest cell identification.
- Battery health classification.
- State of Charge (SOC) estimation.

- Runtime operating modes:
  - Normal
  - Degraded
  - Failsafe
  - Shutdown
- Intelligent fault detection:
  - Weak Cell Detection
  - Overvoltage Detection
  - Sensor Fault Detection
  - Frozen ADC Detection
  - Relay Mismatch Detection
  - Rapid Voltage Change Detection
  - Invalid ADC Reading Detection
- Automatic relay cutoff.
- Buzzer warning system.
- LCD diagnostic display with multiple rotating screens.
- Real-time cloud telemetry using Blynk.
- Fault history logging.
- Risk level estimation.
- Operator recommendations.
- Voltage trend visualization.

Working Methodology

The ESP32 samples ADC values, computes pack parameters, evaluates safety conditions, updates the LCD, controls relay/buzzer, and uploads significant state changes to Blynk.



Results

The developed Battery Intelligence System successfully demonstrated reliable monitoring and protection of a simulated four-cell battery pack.

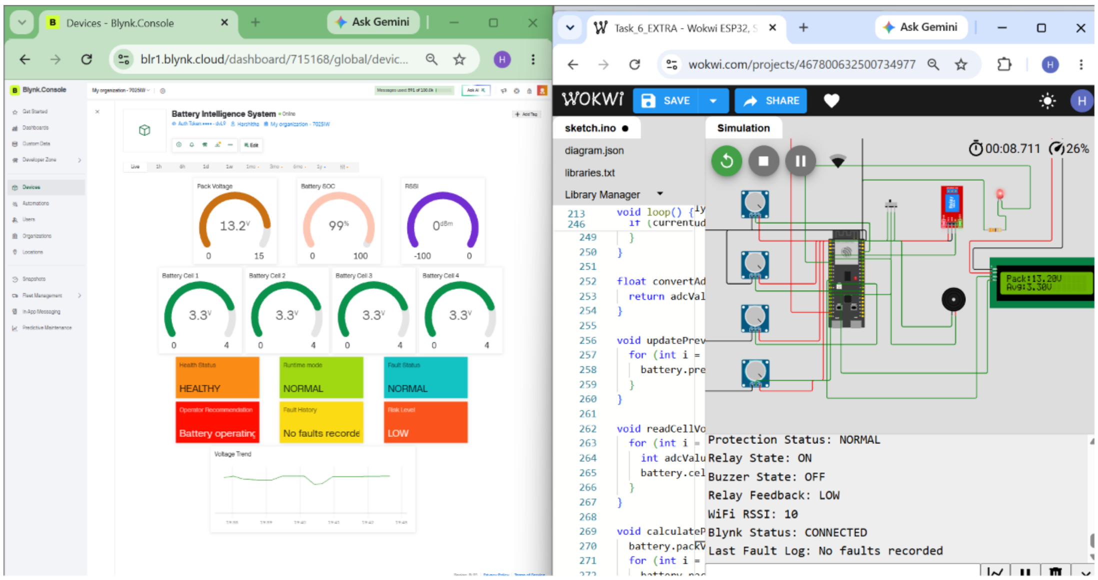
The implemented system successfully performed:

- Real-time battery monitoring.
- Accurate voltage calculations.
- Intelligent battery health classification.
- Automatic fault detection.
- Runtime mode transitions.
- Automatic relay protection.

- Cloud telemetry using Blynk.
- Live LCD diagnostics.
- Battery State of Charge estimation.

- Intelligent operator recommendations.
- Fault history management.
- Voltage trend visualization.

The project operated successfully in Healthy, Fault, Recovery, and Shutdown conditions during testing.



Conclusion

The developed Battery Intelligence System successfully combines embedded systems and IoT concepts into a reliable battery monitoring prototype suitable for learning and demonstration.